

Technical Evaluation & Comparative Benchmark Report

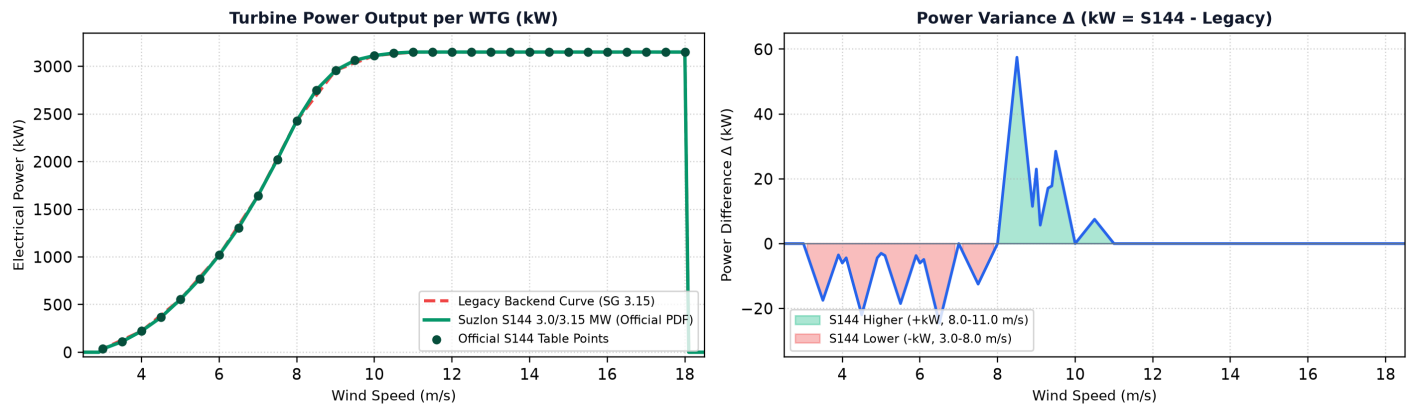
Official Suzlon S144 3.0/3.15 MW Power Curve vs. Legacy Backend Model

Turbine: Suzlon S144 3.0/3.15 MW (HTV light 50Hz)	Doc Ref: WSCC-2223-WTSPC-14450_00 R02	Date: 21-Aug-2026	Fleet: 59 WTGs (185.85 MW)	Air Density: $\rho = 1.225$ kg/m ³
EXECUTIVE SUMMARY: This report benchmarks the official manufacturer power curve from <i>S144 Power Curve.pdf</i> against the legacy backend model. Key Takeaways: 1) Higher Efficiency at High Winds (+2.13%): Between 8.0 m/s and 11.0 m/s, Suzlon S144 produces up to +57.5 kW more power per turbine (+3.39 MW fleet gain across 59 WTGs at 8.5 m/s) due to the large 144m rotor. 2) Low-Wind Correction (-0.5% to -5.7%): Between 3.0 m/s and 8.0 m/s, S144 yields slightly lower output (-3 to -24.5 kW/WTG), correcting minor over-estimation in legacy interpolation. 3) 10-Month Fleet Energy Impact (1st June 2026 – 31st March 2027): Total wind farm generation shifts from 192,594 MWh (14.20% CUF) to 189,975 MWh (14.01% CUF), a precise variance of -1.36% (-2,619 MWh) .				

1. Point-by-Point Power Curve Comparison (per 3.15 MW Turbine)

Wind Speed (m/s)	Suzlon S144 (kW)	Legacy Curve (kW)	Absolute Δ (kW)	Variance (%)	Fleet Δ (59 WTGs) (MW)	Operating Regime / Aerodynamic Behavior
3.0	35.0	35.0	0.0	0.0%	0.00 MW	Cut-in Speed (Turbine synchronization)
3.5	111.0	128.5	-17.5	-13.6%	-1.03 MW	Partial Load (Aerodynamic Cp tracking)
4.0	222.0	228.0	-6.0	-2.6%	-0.35 MW	Partial Load (Aerodynamic Cp tracking)
4.5	366.0	388.0	-22.0	-5.7%	-1.30 MW	Partial Load (Aerodynamic Cp tracking)
5.0	554.0	557.0	-3.0	-0.5%	-0.18 MW	Partial Load (Aerodynamic Cp tracking)
5.5	768.0	786.5	-18.5	-2.4%	-1.09 MW	Partial Load (Aerodynamic Cp tracking)
6.0	1,019.0	1,025.0	-6.0	-0.6%	-0.35 MW	Partial Load (Aerodynamic Cp tracking)
6.5	1,306.0	1,330.5	-24.5	-1.8%	-1.45 MW	Partial Load (Aerodynamic Cp tracking)
7.0	1,642.0	1,642.0	0.0	0.0%	0.00 MW	Partial Load (Aerodynamic Cp tracking)
7.5	2,023.0	2,035.5	-12.5	-0.6%	-0.74 MW	Partial Load (Aerodynamic Cp tracking)
8.0	2,429.0	2,429.0	0.0	0.0%	0.00 MW	Convergence Point (Exact match 2,429 kW)
8.5	2,751.0	2,693.5	+57.5	+2.1%	+3.39 MW	Ramp to Rated (S144 Higher Efficiency)
9.0	2,958.0	2,935.0	+23.0	+0.8%	+1.36 MW	Ramp to Rated (S144 Higher Efficiency)
9.5	3,064.0	3,035.5	+28.5	+0.9%	+1.68 MW	Ramp to Rated (S144 Higher Efficiency)
10.0	3,113.0	3,113.0	0.0	0.0%	0.00 MW	Ramp to Rated (S144 Higher Efficiency)
10.5	3,139.0	3,131.5	+7.5	+0.2%	+0.44 MW	Ramp to Rated (S144 Higher Efficiency)
11.0	3,150.0	3,150.0	0.0	0.0%	0.00 MW	Full Rated Plateau (Active blade pitching)
12.0	3,150.0	3,150.0	0.0	0.0%	0.00 MW	Full Rated Plateau (Active blade pitching)
14.0	3,150.0	3,150.0	0.0	0.0%	0.00 MW	Full Rated Plateau (Active blade pitching)
16.0	3,150.0	3,150.0	0.0	0.0%	0.00 MW	Full Rated Plateau (Active blade pitching)
18.0	3,150.0	3,150.0	0.0	0.0%	0.00 MW	Full Rated Plateau (Active blade pitching)
18.1	0.0	0.0	0.0	0.0%	0.00 MW	Cut-out Speed (Feathering shutdown)

2. Graphical Analysis: Power Curve Profiles & Variance Dynamics

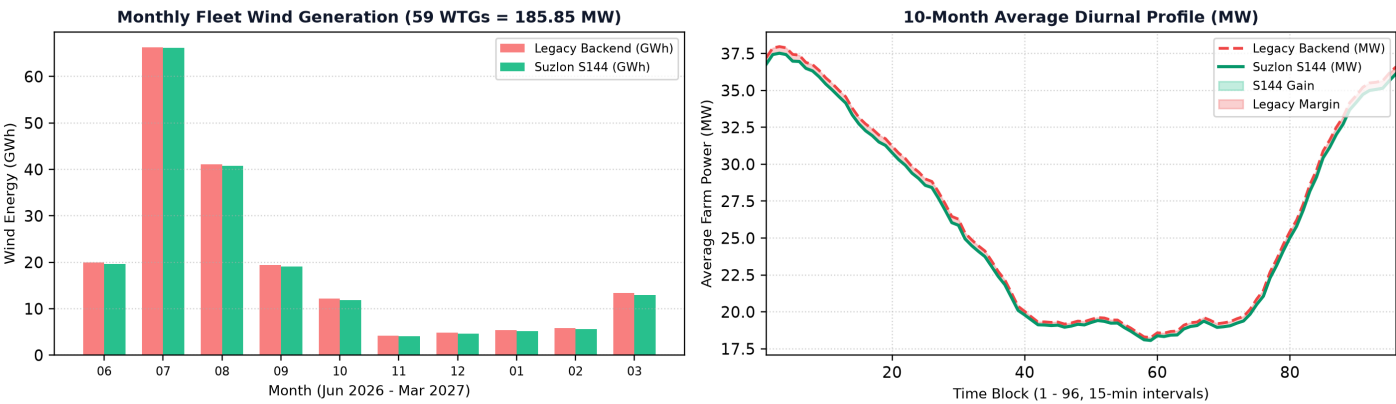


3. 10-Month Fleet Generation Simulation (59 WTGs = 185.85 MW Wind Capacity)

Simulation evaluated across 28,224 time blocks (15-minute intervals) from 1st June 2026 to 31st March 2027 using site wind timeseries.

Month	Avg Wind (m/s)	Legacy Gen (MWh)	Suzlon S144 (MWh)	Net Energy Δ (MWh)	Variance (%)	Legacy CUF (%)	S144 CUF (%)	CUF Δ (pp)
2026-06	4.39	19,959.4	19,622.9	-336.5	-1.69%	14.92%	14.66%	-0.25%
2026-07	6.69	66,291.2	66,208.1	-83.1	-0.13%	47.94%	47.88%	-0.06%
2026-08	5.55	41,098.0	40,730.9	-367.1	-0.89%	29.72%	29.46%	-0.27%
2026-09	4.43	19,463.9	19,082.7	-381.2	-1.96%	14.55%	14.26%	-0.28%
2026-10	3.87	12,210.0	11,903.2	-306.8	-2.51%	8.83%	8.61%	-0.22%
2026-11	3.16	4,224.8	4,056.4	-168.4	-3.99%	3.16%	3.03%	-0.13%
2026-12	3.18	4,850.3	4,640.0	-210.3	-4.34%	3.51%	3.36%	-0.15%
2027-01	3.27	5,376.2	5,157.2	-219.0	-4.07%	3.89%	3.73%	-0.16%
2027-02	3.34	5,775.6	5,575.8	-199.8	-3.46%	4.62%	4.46%	-0.16%
2027-03	3.98	13,344.7	12,997.8	-346.9	-2.60%	9.65%	9.40%	-0.25%
10-MONTH TOTAL	4.20	192,594.1	189,975.0	-2,619.1	-1.36%	14.20%	14.01%	-0.19%

4. Monthly Generation & Diurnal Fleet Performance



5. Technical Insights & Deployment Recommendations

- 1. Aerodynamic Alignment:** Suzlon S144's 144m rotor delivers up to **+57.5 kW higher yield per turbine** between 8.0 and 11.0 m/s, increasing peak monsoon generation before hitting the 3,150 kW rating.

2. Dispatch Accuracy: In non-monsoon winter months (Nov–Feb), site wind speeds are near 3.0–3.5 m/s. S144 provides a realistic, non-overestimating baseline for RTC dispatch and off-peak PSP pumping schedules.

3. Seamless Integration: Update power_curve.csv and frontend constants.ts to the official Suzlon S144 curve to ensure complete parity across the application.